

Assessment of Agility in Software Organizations with a Web-Based Agility Assessment Tool*

Onat Ege Adalı
Information Systems,
TU/e
Eindhoven, Netherlands
o.e.adali@tue.nl

Özden Özcan-Top
Regulated Software Research
Centre and Lero, DKIT
Dundalk, Ireland
ozden.ozcantop@dkit.ie

Onur Demirörs
Information Systems Dept.,
METU & UNSW
Ankara, Turkey & Sydney,
Australia
demirors@metu.edu.tr

Abstract— Agile software development approaches have emerged to overcome the inherent challenges of the traditional software development methodologies. With that in mind, they offer a set of simple yet effective practices based on the values and supporting principles of the Agile Manifesto. Today, more and more organizations are practicing Agile but transitioning to Agile is not as straightforward as it might seem. The benefits afforded by the agile values, principles and practices are not fully obtained and adapters struggle to become more mature; i.e. increase their Agility. For, adapters to address this problem, agility assessments have been designed to be aware of the situation of their transition and make sure that they are heading in the right direction. However, these assessments still require significant amount of time and effort to apply. To overcome this problem, we have developed a web-based agility assessment tool which is a part of an integrated agility assessment approach. The approach consists of a structured and comprehensive agility assessment model (AgilityMod), an exemplar agility assessment process and a web based agility assessment tool (AssessAgility). This research focuses on the final component of the approach: AssessAgility tool and presents a multiple case study that was conducted to evaluate tool's capabilities in terms of providing guidance, automation and reducing the effort spent on the assessments.

Keywords—agility assessment; agile software development; tool; automated; web-based

I. INTRODUCTION

According to the 10th Annual State of the Agile Survey [1] 95% of the total of 3880 respondents said that their organizations practice “Agile” in software development. The survey also states that “agile software development has grown increasingly popular over the last decade”. Agile Methods offer a set of practices based on the four Agile values and twelve supporting principles defined in the Agile Manifesto [2] and they provide the flexibility to adapt these principles and practices, based on the culture, values, and needs of an organization or a team [3].

* This research is partially supported by Science Foundation Ireland Research Centres Programme, through Lero - the Irish Software Research Centre (<http://www.lero.ie>), grant 10/CE/I1855 & 13/RC/2094 and partially supported by the Scientific and Technological Research Council of Turkey (TÜBİTAK), grant number 113E528.

However, agile adaptation within an organization is not easy and straightforward. The challenges in relation to adaptation include misinterpretation of Agile concepts, focusing only on implementing agile practices while ignoring agile values and principles (*doing agile vs being agile*), and sticking to implementing a not well suited practice rather than quickly trying, observing and discarding that practice and performing the adaptation in a continuous way. In addition to these, “Agile” could be used as an excuse for being undisciplined.

Agility assessment, which is a crucial instrument to understand the status of an adapted Agile Method in an organization, enable adopters to find out if the applied methodology has met its intended purpose and to determine improvement areas for improved agility.

The approach we propose for agility assessment include three components: a structured and method-independent *agility assessment reference model (AgilityMod)* [4], an *exemplar agility assessment process* having the definitions and guidance to conduct assessments with AgilityMod and a *tool (AssessAgility)* that automates and guides the assessment process based on the exemplar assessment process.

In this study, we focus on presenting the third component of the approach; the web based agility assessment tool. We have performed a multiple case study to evaluate the tool's capabilities in terms of providing guidance, automation and reducing the effort spent on the assessments.

The *AssessAgility tool* has been developed to improve the efficiency in agility assessments performed with AgilityMod with its superior functions compared to existing agility assessment tools. The tool supported approach proposed in this study is able to provide guidance and automation in planning, conducting, and reporting stages.

The rest of this paper is organized as follows: In Section II, we present the study that was performed to specify and evaluate the existing agility assessment tools, the minimum criteria that should be in an assessment tool and related research on agility assessment. In Section III, the AssessAgility: Web Based Agility Assessment Tool is presented. In section IV, we present the multiple case study that we have conducted to evaluate our

approach. In section V, we provide the conclusion and finally in section VI we provide directions for the future research.

II. LITERATURE REVIEW

A. Agility Assessment Tools and Evaluation

Agility assessment tools exist in the literature could be classified into templates and software programs that are developed with the purpose of automating and guiding the agility assessment process. The main intent for the development of these tools is to measure to what extent a software company has succeeded to adapt these agile methods in its software development process [5]. In our previous study [6], we have identified 46 agility assessment tools exist in the literature. That study included evaluation of 14 of the 46 tools which met the inclusion criteria with a multiple case study based on nine criteria. These nine criteria which should be in an agility assessment tool, were deducted based on the set of prominent features of the currently available agility assessment tools and the opinions of the experts. The criteria were: coverage, availability, guidance capability, assessment recording, automated reporting, comparability, different modes of usage, different scopes and extensibility. The brief description of each criterion is given below:

- Coverage: Ability to investigate and cover 12 Agile Principles stated in the Agile Manifesto [2].
- Availability: Ability to be universally reachable for researchers and practitioners.
- Guidance Capability: Ability to provide guidance to users before, during and after the assessment.
- Assessment Recording: Ability to record agility assessment findings and the resulting reports for further modifications, analysis, and comparison.
- Automated Reporting: Ability to automatically generate reports for the presentation of the results of the performed assessments.
- Comparability: Ability to provide comparison between the reports of previously performed assessments.
- Different Modes of Usage: Ability to support different usage mods for individuals, multiple users, and parallel assessments.
- Different Scopes: Ability to perform assessments on project, team, and/or organizational levels.
- Extensibility: Ability to provide extensibility to meet emerging needs of different types of assessment contexts.

The purpose of the multiple case study was to identify sufficiency of the current agility assessment tools to meet the expected criteria. We have evaluated these tools by implementing them in a software development organization that employs an established agile software development method, namely Scrum. Fig. 6 below shows the results of the multiple case study. Some criteria were rated by using a four-point ordinal (NA: Not Achieved-PA: Partially Achieved - LA: Largely Achieved - FA: Fully Achieved) scale and some rated by using a dichotomous (yes/no) scale.

The details of this study could be found at [6]. Another observation of the study was that the tools' lack of support for the important steps of the assessment process such as planning and data validation. We also observed that the majority of the tools do not provide levels of agility or issues and possible improvement areas to achieve agility better. Our final observation shows that these tools rely on specific set of agile practices (such as daily stand-up meetings, pair programming etc.) to indicate agile adoption. While these practices are crucial for implementation of specific agile methods, the mere absence or presence of these practices are not sufficient to indicate the success of the agile adoption. All in all, the literature review and evaluation of the agility assessment tools revealed the need for an agility assessment tool that is capable of meeting the specified criteria above, provides guidance in agility assessment and automation for the assessment process as a whole. It's expected that such a tool provides assessment results that includes an indication of agility levels, the possible improvement areas towards agility by relying on a comprehensive agility assessment model rather than a limited set of agile practices.

B. Related Research on Agility Assessment Tools

It was observed that only a few studies have been conducted on comparison of the agility assessment tools. Below we also included studies in relation to agile maturity models.

There are about forty models related to agile maturity, published in academic journals and conferences and on web pages [7, 8]. Schweigert *et al.* [7] argue that these models do not quantify the real agility and support guidance. Instead, they check for the implementation of some specific agile practices.

In one of our previous studies [9] five of the most frequently referenced agile maturity models were applied in an organization and evaluated based upon the six quality criteria: *fitness for purpose, completeness, definition of agile levels, objectivity, correctness and consistency*. The results of the study indicated that none of these models satisfies all the expected criteria and is needed to be improved in terms of scope, definitions of agility levels and objectivity.

Soundararajan and Arthur [5] have reviewed Comparative Agility [6], and Thoughtworks Agile Assessment [7] tools. They compared the tools according to their ability to measure agility. Soundararajan and Arthur concluded that these agility assessment tools "primarily focus on the presence or absence of practices" and they have stated that presence or absence of practices is not sufficient to "indicate an organization's ability to implement an agile method or the effectiveness of an agile method" [5].

In another study [8] Gandomani and Nafchi again reviewed Comparative Agility and Thoughtworks Agile Assessment Tools. They have agreed on the findings of Soundararajan and Arthur on Comparative Agility and they added that the Thoughtworks Assessment Tool does not include the identification of effectiveness of the Agile methods and twenty practices included in the tool are not sufficient.

III. ASSESSAGILITY: WEB BASED AGILITY ASSESSMENT TOOL

The tool AssessAgility is built to provide automation and guidance for the AgilityMod by adapting the accompanying exemplar agility assessment process. The main objective of the tool is to reduce the amount of time spent for the assessment process. The initial requirements of the tool are based on the nine expected criteria that is described in Section II.

Before delving into the details of the AssessAgility, we briefly present the agility assessment reference model, AgilityMod.

A. AgilityMod--An Agility Assessment Reference Model

The web based agility assessment tool and the exemplar assessment process described in this paper have built upon the Agility Assessment Reference Model: AgilityMod that was developed by two of the authors previously [4].

AgilityMod enables assessment of a project's agility based on defined specific practices and agility practices. It's meta-model structure is defined in accordance with ISO/IEC 15504-Process Assessment Model [10-12].

The model mainly consists of two dimensions: Aspect dimension and Agility Dimension which can be seen in Fig. 1.

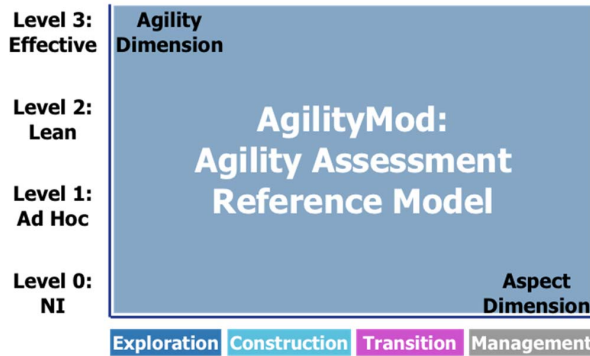


Fig. 1. AgilityMod's Structure.

Aspects are defined as "Exploration, Construction, Transition and Management" which are derived from agile processes and practices. In the other dimension, agility of an aspect is measured with a four-point ordinal scale which enables the agility to be assessed at "Not Implemented", "Ad-Hoc", "Lean" and "Effective" levels. The Aspects are modularization of agile processes and practices that are integrated under meaningful and agile compatible abstract definitions. When an aspect progresses from the bottom level: "Not Implemented" to the top level: "Effective", its conformance to agile values and principles increases. Details on AgilityMod could be found in [13].

B. The Design and Features of the Tool

The tool is designed as a web based cloud application and developed within four two-week long time-boxed iterations. We employed use case based modelling for the elaboration of the requirements and used Model View Controller (MVC) pattern for the architecture of the tool. The development language for the application was mainly C#. The Microsoft Visual Studio was used as an Integrated Development

Environment (IDE). Furthermore, Javascript, Bootstrap Framework, and SQL Server Compact (SQL CE) technologies were used for the development. The tool is publicly available and can be seen at assessagility.azurewebsites.net.

In terms of the graphical user interface the tool provides distinct sections for each role and responsibility specified in the exemplar agility assessment process. The sections include pages for users to perform actions and guidance facilities such as notes, information buttons and suggestions are also included in the pages.

For the Lead Assessors, tool provides facilities for the management of teams, assignments, assessments and reports. With the tool, Lead Assessors can create teams, make assessment plans, assign assessments to the teams, observe the progress of the assignments, and generate reports.

For the Assessors, tool provides facilities to perform the assessments, take notes, and upload objective evidences. During the performance of the assessments, tool enables the assessors to record the data that they have collected to specific fields, match the collected data with objective evidences and view the progress and results of the assessments in real time.

IV. MULTIPLE CASE STUDY

The multiple case study was designed with the objectives of determining whether or not the tool meets the expected criteria we have defined, is able to guide and automate the exemplar agility assessment process as a whole and is able to provide efficient usage for performing agility assessments. In addition to that, we requested from the assessment team to evaluate its efficiency and report the strengths and weaknesses that they have identified while performing the assessments.

A. Design of the Case Study

In the light of the objectives given above, we have defined the following research questions:

RQ1: To what extent is the tool sufficient to meet the expected criteria determined after the literature review of this study?

RQ2: To what extent is the tool able to automate and guide the agility assessment process?

RQ3: How efficient is the tool for the assessment process?

RQ4: What are the strengths and weaknesses of the tool?

To increase the reliability of our validation, we have planned to conduct a multiple case study including three cases. Each case study was planned to be conducted on a different project from different business domains and from different types and sizes of organizations. The only strict requirement for projects was that they should be carried out in an Agile setting. In the scope of the case studies we planned to conduct an agility assessment covering 4 aspects defined in the reference model for each project. Each assessment was planned to be performed in conformance with the exemplar assessment process defined in the Section IV. To perform the assessments, we aimed to form an assessment team that consists of two assessors and one assessment leader. These experts were planned to be assigned to relevant roles, according to their experience in the field. We also planned to pick software process improvement experts to increase objectivity.

1) *Data Collection Strategy*: In the Data Collection phase of the assessment process we planned to conduct structured interviews with the aspect owners of the projects. The aspect owners that were planned to be interviewed include at least one requirements engineer, one product owner, one developer, one tester and one configuration manager. The main reason behind conducting interviews with different roles was to get direct responses from relevant practitioners. The responses given in the interviews were recorded to the related fields provided in the tool and in a database located on a cloud server.

2) *Validation Strategy*: The generated reports were discussed with the assessment team members to discuss the results of the assessment.

B. Conduct

As stated in the case study design we performed three case studies on three projects in different business domains and from different types and sizes of organizations. An assessment team was formed with an assessment leader who has 5 years of experience as an SPI expert and two assessors one of them having one and a half years of experience and the other one having one year of experience as an SPI consultant. Before the conduct of the case studies, each team member in the assessment team completed a manual assessment without the tool to be able to grasp the reference model and the exemplar agility assessment process. The total time for each assessor to complete the manual assessment was tracked and recorded to compare with the assessments performed with the tool. The team followed the steps of the exemplar assessment process given in the Fig. 3 by using the tool. Then, an assessment report for each case was obtained and discussed with the assessment participants and their teams for validation. The details and findings of each case are given in the next section. After the interviews, we also asked assessors to answer the following questions.

TABLE I. CASE STUDY EVALUATION FORM

ID	Question	Related RQ
Q1	Please rate the tool with the evaluation questionnaire provided.	RQ1
Q2	Does the tool cover the assessment process as a whole? If not, please describe the missing parts of the process.	RQ2
Q3	Does the tool guide the assessment process as a whole? If you required additional guidance about the assessment process, please describe the parts where you needed guidance?	RQ2
Q4	Please rate the usability of the tool with the questionnaire provided.	RQ3
Q5	Does the tool reduce the effort spent for performing assessments? Could you please describe the difference?	RQ3
Q6	Could you please describe the strengths and weaknesses of the tool?	RQ4

For question 1 we have used Evaluation Questionnaire, for question 2 we have used USE Questionnaire: Usefulness, Satisfaction, and Ease of Use based on [16].

1) Case Study 1:

In the scope of the Case Study 1, the assessment is performed through a total of three-hour interview session with two team

leaders. Each team leader is interviewed by one assessor and whole interview is completed within total of 12 person-hours. The details of the case study are given in the table below:

TABLE II. CASE STUDY I BIO

Organization ID:	Organization GS
Size of the Organization:	Medium SME (100-250 Employees)
Agile Experience:	>3 Years
Organization Bio:	Organization GS is data analytics service provider which offers various SAAS based web applications for its customers. Currently it has 20 customers constantly using their various services such as data visualization, data processing, decision support, social network analysis, marketing analytics and customer behavior analysis. The organization decided to transit to Agile about four years ago with the main objectives of reducing time to market and enhancing software maintainability. All the current projects are applying Agile and Scrum is used throughout the projects with some customizations.
Project ID:	Project 1
Project Bio:	The project assessed in the scope of the study was a new development project which involves development of an SNA Module for an existing web service framework. It has several interfaces to other modules such as data visualization, forecasting, and prediction. The framework which the developed SNA Module will be a part of is a group of web application services which provides continuous service delivery, data protection, and prediction.
Project Team Bio:	The project was carried out by 2 separate teams with a total of 20 team members. The distribution of the roles is given below: • 2 Team Leaders • 7 Software Developers • 2 Software Architects • 2 Data Visualization Analysts • 5 Data Analysts • 2 Testers

2) Case Study 2:

The assessment for the Case Study 2 is performed through a total of five-hour interview session with a business analyst and a software developer. Each person is interviewed by one assessor and the whole interview is completed within a total of 15 person-hours. The details of the Case Study 2 are given in the table below:

TABLE III. CASE STUDY II BIO

Organization ID:	Organization TX
Size of the Organization:	Small SME (25-100 Employees)
Agile Experience:	1,5 Years
Organization Bio:	Organization TX is an IT firm that mostly provides customizable IOT services for on-site business users. Their customer portfolio mainly consists of rent-a-car firms, event holders, fair and conference venues, universities, shopping malls, art galleries and museums. Their current services include indoor localization services that are based on iBeacons, vehicle tracking, and remote device management. The organization decided to adopt Agile methodologies 2 years ago and since then they are applying a Scrum XP Hybrid for all of their projects.
Project ID:	Project 2
Project Bio:	The project assessed in the scope of the study was an existing solution implementation case which involved usage of an existing platform. It involved the

	implementation of remote device management and communication services to a manufacturing factory.
Project Team Bio:	The project was carried out by a team that consists of 11 members. The distribution of the roles is given below: • 2 Business Analysts • 1 Business Owner • 2 Electrical Engineers • 5 Software Developers • 1 Tester

3) Case Study 3:

The assessment for the Case Study 3 is performed through a total of 2,5-hour interview session with the game leader and a software developer. Each person is interviewed by one of the assessor and whole interview is completed within total of 10 person-hours. The details of the Case Study 3 are given in the table below:

TABLE IV. CASE STUDY III BIO

Organization ID:	Organization BV
Size of the Organization:	Small SME (25-100 Employees)
Agile Experience:	2 Years
Organization Bio:	Organization BV is an indie game development studio that develops cross platform game applications. In total the studio has released 18 mobile games in different genres for Android and iOS mobile platforms. The Agile is the agreed software development method for the studio since the beginning and a Kanban based customized method is used for the projects.
Project ID:	Project 3
Project Bio:	The project assessed in the scope of the study was a new development project that involves development of a third person action adventure game. The game is based on the neural network AI technologies and has the ability to adapt the main scenario to player's gaming style. The project was developed with the Unity Game Engine.
Project Team Bio:	The project was carried out by a team that consists of 9 members. The distribution of the roles is given below: • 1 Game Leader • 1 3D Storyboard Artist • 2 Graphic Designers • 3 Software Developers • 1 Voice Engineer

C. Findings of the Case Study

The findings on the tool and agility levels of the organizations are discussed below.

1) Findings on RQ1: To what extent is the tool sufficient to meet the expected criteria?

The ratings given by assessors and the lead assessor are given in the table below. Considering the ratings given, it can be said that the AssessAgility has fully complied with the 8 out of 9 expected criteria: Coverage, Availability, Guidance Capability, Assessment Recording, Automated Reporting, Comparability, and Extensibility. The only criterion that could not be fully complied was Different Scopes.

TABLE V. FINDINGS ON RQ1

Criteria	Lead Assessor	Assessor 1	Assessor 2
Coverage	FA	FA	FA
Availability	Web-Based	Web-Based	Web-Based
Guidance Capability	FA	FA	FA

Assessment Recording	Yes	Yes	Yes
Automated Reporting	Yes	Yes	Yes
Comparability	Yes	Yes	Yes
Different Scopes	LA	LA	LA
Extensibility	Yes	Yes	Yes

2) Findings on RQ2: To what extent is the tool able to automate and guide the assessment process?

For the RQ2, we request from assessment team members to rate the automation (facilitation of the execution of the assessment steps) and guidance capability of the tool for each phase of the exemplar assessment process.

TABLE VI. FINDINGS ON RQ2

Phase	Lead Assessor		Assessor 1		Assessor 2	
	Automation	Guidance	Automation	Guidance	Automation	Guidance
Planning	FA	FA	FA	FA	FA	FA
Data Collection	FA	FA	FA	FA	FA	FA
Validation and Agility Level Determination	FA	FA	FA	FA	FA	FA
Reporting	FA	FA	FA	FA	FA	FA

3) Findings on RQ3: How efficient is the tool for the assessment process?

As shown in the Table I, for the RQ3 we asked questions: Q4, and Q5 to the assessors. The findings on each question are given below:

a) Q4: Please rate the usability of the tool with the questionnaire provided.

TABLE VII. The ratings given by each team member for each area of the questionnaire is given below:

TABLE VIII. LEAD ASSESSOR USE QUESTIONNAIRE RATINGS

Usefulness: 53/56	Ease of Use: 75/77
Ease of Learning: 28/28	Satisfaction: 49/49

TABLE IX. ASSESSOR 1 USE QUESTIONNAIRE RATINGS

Usefulness: 54/56	Ease of Use: 72/77
Ease of Learning: 27/28	Satisfaction: 49/49

TABLE X. ASSESSOR 2 USE QUESTIONNAIRE RATINGS

Usefulness: 55/56	Ease of Use: 77/77
Ease of Learning: 28/28	Satisfaction: 49/49

The questions in the USE questionnaire are constructed as seven-point Likert rating scale, a rating of seven means strong agreement and a rating of zero means strong disagreement. The ratings suggest that assessment team strongly agrees to the positive statements in the questionnaire therefore according to

the USE Questionnaire, AssessAgility's usability is high in terms of the questionnaire's three dimensions: Usefulness, Satisfaction, and Ease of Use.

b) Q5: Does the tool reduce the effort spent for performing assessments? Could you please describe the difference?

For this question, we requested from the assessment team members to perform a manual assessment for a pilot case and we wanted them to record the effort that they have spent for the manual assessment and compare it to the average effort that they have spent for the assessments performed for each case in the multiple case study. The comparison was made on the effort spent by each team member for the phases that they were responsible for. The comparison is given in the Fig. 2.

The combined average effort decrease for the three team members is 40%.

Lead Assessor	Assessor 1	Assessor 2
Based on the effort spent for Planning, Validation and Agility Level Determination and Reporting	Based on the effort spent for Data Collection and Reporting	Based on the effort spent for Data Collection and Reporting
Pilot: 3,6 prs/hrs	Pilot: 6 prs/hrs	Pilot: 5 prs/hrs
Avg. of Cases: 1,75 prs/hrs	Avg. of Cases: 3,5 prs/hrs	Avg. of Cases: 3,5 prs/hrs
50% Decrease	40% Decrease	30% Decrease

Fig. 2. The Comparison of Efforts Spent by Team Members for the Manual and Tool Supported Assessments.

4) Findings on RQ4: What are the strengths and weaknesses of the tool?

The answers from all three team members are given below:

a) Strengths:

- Ease of management of assessments
- Ease of assessment performance
- Ease of use and understanding
- Clear, responsive, and useful user interface
- Guidance for the assessment process
- Automated reports
- Support for parallel and distributed assessments
- Easy data and evidence collection

b) Weaknesses:

- Tool requires an instant messaging facility which could ease the coordination between distributed team members
- Notification feature to let users about the assignments and their status

D. Findings on the Agility Levels of the Organizations

1) Organization 1

We assessed the Organization GS over the Project 1 which is a new development project that involves development of an SNA Module for an existing web service framework. The achieved agility levels for the project can be seen on the Fig. 3 below:

The results show that all aspects are at the Level 3: Effective however our findings about the practices suggested that for Construction and Transition Aspects, specific training

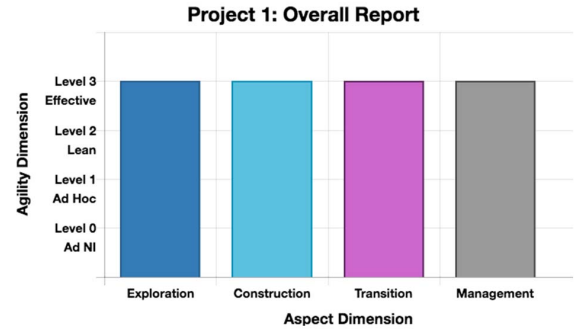


Fig. 3. Achieved Agility Levels of Project 1

mechanisms can be employed to encourage people to learn. And again, for the Transition Aspect, we proposed analysis of collected metrics for improvement of the aspect.

2) Organization 2

We assessed the Organization TX over the Project 2 which is an existing solution implementation case that involved usage of an existing platform. The achieved agility levels for the project can be seen on the Fig. 4 below:

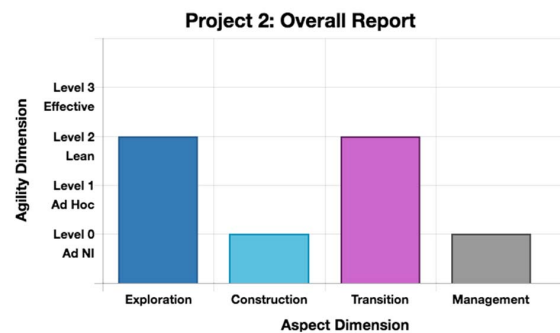


Fig. 4. Achieved Agility Levels of Project 2

The results show that Exploration and Transition Aspects are at the Level 2: Lean and Construction and Management Aspects are at Level: 1 Ad-Hoc. Our findings about the practices suggested that Agile focused learning mechanisms were missing that will help to train people on the Exploration and Transition Aspects. For the Construction Aspect, we observed that developer level tests were not consistent and for the Management Aspect we observed some traditional project management approaches such as command and control approach that were not aligned with Agile Methodologies.

3) Organization 3

We assessed the Organization BV over the Project 3 which new development project that involves development of a third person action adventure game. The achieved agility levels for the project can be seen on Fig. 5 the below:

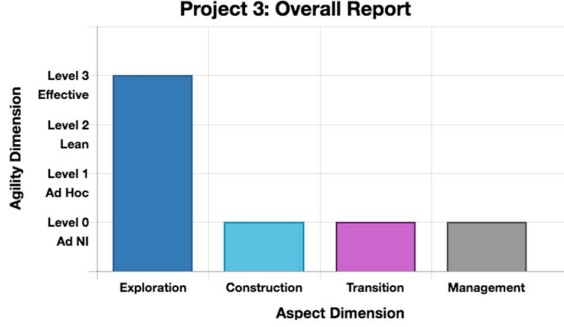


Fig. 5. Achieved Agility Levels of Project 3

The results show that Exploration Aspect is at the Level 3: Effective and Construction, Transition, and Management Aspects are at Level: 0 Not Implemented. The findings on the Exploration Aspect revealed that the requirements for the game applications were developed in-house by the project teams and these teams were perfectly capable of managing the conflicts on requirements. For the Construction Aspect, our findings on the practices suggested that metrics were not collected. For the Transition Aspect, we observed that the deployment was randomly done and correctness of the software was not ensured. Finally, for the Management Aspect our findings suggested that, project plans were not adapted to change and risk mitigation was not performed.

VALIDITY THREATS

Due to the fact that multiple case study approach employs qualitative data, it's prone to validity threats. According to Yin [17]; the quality of case studies can be evaluated by four tests: construct validity, internal validity, external validity and reliability. Below, the techniques employed to address each test is given with details:

Construct validity concerns with how well an experiment aligns to its claims. For ensuring construct validity, we gathered evidences from multiple sources: documentation, interviews and participant observation. These sources are three of the total of six sources that Yin [17] suggested. Furthermore, the evidence was stored with the assessment reports in the database to ensure the construct validity by storing and then we gave access to experts to get feedback about the validity of the case study.

Internal validity concerns with the achievement of casual conclusion by minimizing the systematic error or bias. In order to ensure internal validity, we employed explanation building technique suggested by Yin [17]. We discussed the assessment results within reviews held with the attendance of every team member in the assessed projects. We presented evidence and explained the relationship between each evidence and the

obtained practice rating. Each rating and evidence was discussed with team members to avoid any bias in the results. External validity concerns with the generalization of results to other contexts. In order to ensure external validity, we performed assessments with three different organizations working in three different business domains, and developing three different types of software. (See Tables II, III, and IV for the information on cases.) Furthermore, for each assessment two assessors gathered separate evidences and these evidences are reviewed by lead assessor. Therefore, the tool was used in three different settings with three different people and we believe that this will increase the generalizability. Reliability concerns with the repeatability of the study by eliminating errors and biases. To ensure reliability we employed multiple case study approach and we replicated the results by performing assessments on same project (or same setting) with two different assessors.

V. CONCLUSION

The main purpose of this research was to develop an agility assessment tool which will reduce the amount of time and effort spent for agility assessments for organizations to continuously evaluate their Agile Adoptions to increase their Agile Maturity. To evaluate the tool against this purpose, we designed a multiple case study. The objectives of the multiple case study were to evaluate the tool's ability to meet the expected criteria and to learn whether or not it has the ability to reduce the effort and time spent for the agility assessment process by fully automating and guiding it.

In terms of meeting the expected criteria, Analysis of the case study results, revealed that the tool is able to fully achieve 8 out of 9 expected criteria. The comparison of the tool's evaluation against other available tools is given in the Fig. 6.

#	Tool/Quality Criteria	Coverage	Availability	Guidance Capability	Assessment Recording	Automated Reporting	Comparability	Different Modes of Usage	Different Scopes	Extensibility
1	Agile Assessment	FA	Not Web-Based	PA	Yes	Yes	No	NA	NA	No
2	Agile Enterprise Survey	LA	Web-Based	PA	No	No	No	NA	NA	No
3	Agile Health Dashboard	LA	Not Web-Based	FA	Yes	Yes	Yes	NA	NA	Yes
4	Agile Journey Index	FA	Not Web-Based	FA	Yes	Yes	Yes	NA	NA	No
5	Agile Process Assessment Tool	LA	Not Web-Based	PA	Yes	Yes	No	NA	NA	No
6	Agile Self Assessment	FA	Web-Based	LA	Yes	Yes	No	LA	NA	No
7	Agility Questionnaire	LA	Not Web-Based	LA	Yes	Yes	No	NA	FA	Yes
8	Comparative Agility	FA	Web-Based	FA	Yes	Yes	Yes	NA	NA	Yes
9	Depth of Kanban	LA	Not Web-Based	PA	N/A	No	No	NA	NA	Yes
10	Enterprise Agility Maturity Matrix	FA	Not Web-Based	LA	Yes	Yes	No	NA	LA	No
11	GSPA	FA	Not Web-Based	LA	Yes	Yes	No	LA	LA	Yes
12	IBM DevOps Practices Self Assessment	LA	Web-Based	LA	Yes	Yes	No	NA	NA	No
13	Open Assessments	FA	Web-Based	FA	No	Yes	No	NA	NA	No
14	Team Metrics	NA	Web-Based	FA	Yes	Yes	Yes	FA	NA	No
15	AssessAgility	FA	Web-Based	FA	Yes	Yes	Yes	FA	LA	Yes

Fig. 6. The results of the evaluation.

In terms of providing guidance and automation for the process, the tool proved to be covering every phase of the exemplar assessment process. On the efficiency, the tool proved to reduce the effort spent on the assessment on average of 30% in contrast to the manual assessment. Finally, the weaknesses and the strengths of the tool are given in the previous section.

In conclusion, this study resulted in development of a web-based online agility assessment tool that has the ability to guide and automate the agility assessment process; including planning, conducting, and reporting phases of the agility assessment process and provides a comprehensive way of assessing Agility without relying on a specific Agile Methodology, an implementation, or a limited set of practices.

VI. FUTURE WORK

For future work, we plan to evaluate the sufficiency of our integrated agility assessment approach as a whole in a series of case studies.

REFERENCES

- [1] VersionOne, "10th Annual State of Agile™ Report," 2015.
- [2] K. Beck, M. Beedle, A. van Bennekum, A. Cockburn, W. Cunningham, M. Fowler, *et al.*, "The Agile Manifesto," 2001.
- [3] S. S. a. J. D. Arthur, "A Structured Framework for Assessing the "Goodness" of Agile Methods," presented at the 18th IEEE International Conference and Workshops on Engineering of Computer-Based Systems, 2011.
- [4] O. Ozcan-Top, "A Reference Model for Software Agility Assessment: AgilityMod " PhD, Middle East Technical University, 2014.
- [5] H. Z. Mina Ziaei Nafchi, Taghi Javdani Gandomani, "On The Current Agile Assessment Methods and Approaches," presented at the Malaysian Software Engineering Conference (MySEC), 2014.
- [6] Ö. Ö.-T. Onat Ege Adalı, Onur Demirörs, "Evaluation of Agility Assessment Tools: A Multiple Case Study," presented at the 16th International SPICE Conference (SPICE 2016), Dublin, Ireland, 2016.
- [7] T. Schweigert, D. Vohwinkel, M. Korsaa, R. Nevalainen, and M. Biro, "Agile Maturity Model: A Synopsis as a First Step to Synthesis," in *Systems, Software and Services Process Improvement*, ed: Springer, 2013, pp. 214-227.
- [8] T. Schweigert, D. Vohwinkel, M. Korsaa, R. Nevalainen, and M. Biro, "Agile maturity model: analysing agile maturity characteristics from the SPICE perspective," *Journal Of Software: Evolution And Process*, 2013.
- [9] Ö. Özcan Top and O. Demirörs, "Assessment of Agile Maturity Models: A Multiple Case Study," in *Software Process Improvement and Capability Determination*, Bremen, Germany, 2013, pp. 130-141.
- [10] I. O. f. Standardization and I. E. Commission, "ISO/IEC 15504-2:2003 Information technology -- Process assessment -- Part 2: Performing an assessment," ed, 2003.
- [11] "ISO/IEC 15504-5:2012 Information technology -- Process assessment -- Part 5: An exemplar software life cycle process assessment model," ed, 2012.
- [12] I. O. f. S. I. a. I. E. C. (IEC), "ISO/IEC 15504-5:2012 Information technology -- Process assessment -- Part 5: An exemplar software life cycle process assessment model," ed, 2012.
- [13] O. Ozcan-Top and O. Demirörs, "A Reference Model for Software Agility Assessment: AgilityMod," in *Software Process Improvement and Capability Determination*. vol. 526, T. Rout, R. V. O'Connor, and A. Dorling, Eds., ed: Springer International Publishing, 2015, pp. 145-158.
- [14] I. O. f. S. I. a. I. E. C. (IEC), "ISO/IEC 15504-3, 2004. ISO/IEC 15504-3:2004 - information technology - process assessment - part 3: guidance on performing an assessment," ed, 2004.
- [15] "Standard CMMI® Appraisal Method for Process Improvement (SCAMPISM) A, Version 1.3: Method Definition Document," Software Engineering Institute 2011.
- [16] A. M. Lund, "Measuring Usability with the USE Questionnaire," *STC Usability SIG Newsletter*, 2001.
- [17] R. K. Yin, *Case Study Research: Design and Methods*, Fifth Edition ed.: SAGE Publications, 2014.